Asymmetric Process-Reward Adaptation for Small-Model Mathematical RL

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Abstract

Final-answer correctness can provide noisy supervision for process reward models (PRMs). We test one loss change: omit positive outcome supervision during offline PRM adaptation while retaining public step labels. Four matched GRPO experiments measure policy-only mathematical accuracy and reward quality under a small-model budget.

1 Research problem and importance

Some answer-correct solutions contain reasoning errors (Zheng et al., 2025). Their positive outcome labels can teach a PRM to reward faulty reasoning. We ask: **does omitting this supervision improve the policy's mathematical accuracy?** We hypothesize that reducing false certification can outweigh losing useful positive supervision; downstream experiments must establish whether this trade-off is favorable.

2 Related work, limitations, and contribution

PRM800K supplies step labels (Lightman et al., 2023); PRIME updates implicit rewards online (Cui et al., 2025); PURE addresses reward hacking through min-form credit assignment (Cheng et al., 2025). LCA learns PRMs from outcomes with softmax-weighted pooling and already studies one-sided label noise (Jia et al., 2026). We use its pooling but test a narrower question: **whether omitting positive outcome supervision, while retaining real step labels, improves downstream GRPO**. The symmetric control is LCA-inspired, not a full reproduction. Our contribution is this controlled supervision change and its policy-level evaluation.

3 Proposed method

Adapt once. Generate four solutions per problem on 512 training problems using the initial policy. Retain complete, unambiguously verified solutions. For step-boundary correctness scores $p_{\phi,t}$, set $e_{\phi,t} = 1 - p_{\phi,t}$. Following LCA,

$$w_t = \frac{\exp(e_{\phi,t}/T)}{\sum_j \exp(e_{\phi,j}/T)}, \qquad E_\phi(\tau) = \sum_t \mathrm{sg}(w_t) e_{\phi,t}, \qquad T = 0.2. \tag{1}$$

Here sg stops gradients through weights. Let $L_{\text{fail}} = \mathbb{E}_{\tau^-}[-\log E_\phi(\tau^-)]$ and $L_{\text{success}} = \mathbb{E}_{\tau^+}[-\log(1 - E_\phi(\tau^+))]$, using separate means over failures/successes. A shared set of 2,048 public positive/negative step labels supplies mean binary cross-entropy L_{step}:

$$L_c = L_{\text{fail}} + cL_{\text{success}} + L_{\text{step}}, \qquad c_C = 1, \quad \boxed{c_D = 0 \text{ (ours)}}. \tag{2}$$

Failures provide weak evidence of an error, without its location. Real step labels discourage blanket rejection; they do not guarantee accurate scores or downstream gains.

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Policy training (continued). For B–D, use the frozen PRM to compute

$$R_i = y_i + 0.2 \min_t p_{\phi,t}(\tau_i), \qquad \hat{A}_i = \frac{R_i - \bar{R}}{\sigma_R + 10^{-4}}. \tag{3}$$

Here $y_i \in \{0, 1\}$ is answer correctness; $\bar{R}, \sigma_R$ are the group mean/std. A uses $R_i = y_i$. The detached sequence advantage applies to all completion tokens in clipped GRPO (Shao et al., 2024). Correct policy rollouts remain in training.

4 Four-group experimental design

Group	Reward / PRM preparation	Controlled comparison
A	Outcome only; no PRM	Outcome-only GRPO baseline
B	Mixed reward; original PRM frozen	B vs. A: existing process reward
C	Mixed reward; adapt with c = 1, then freeze	C vs. B: full adaptation package
D (ours)	Mixed reward; adapt with c = 0, then freeze	**D vs. C: proposed supervision rule**

Implementation. Use Qwen2.5-Math-1.5B as policy and H4 Qwen2.5-Math-1.5B-Instruct-PRM-0.2 as PRM (Hugging Face H4, 2025); LoRA ranks are 16/8. Preserve native step formatting and freeze PRMs throughout RL. TRL/PEFT uses `loss_type="grpo"` and sums rewards before group normalization (Hugging Face, 2026). Truncated/unparseable outputs receive zero reward. Pin all configurations before formal runs.

Data. Reserve 500 MATH training problems for validation; use the remainder for RL/buffer generation (Hendrycks et al., 2021). Deduplicate new training data/anchors against validation, tests, and ProcessBench. PRM800K includes 4,500 original MATH test problems in training (OpenAI, 2023); filtering new data cannot undo inherited PRM exposure. Use **MATH500 as primary test after checkpoint-corpus overlap audit**, GSM8K for transfer (Cobbe et al., 2021), and full MATH only as exposure-aware supplementary evaluation. Unknown pretraining exposure remains a limitation.

Controls and budget. C/D share initialization, batches, optimizer, and updates; only c changes, without re-averaging losses. All arms share policy initialization, prompt order, decoding, and tuning effort. Start with eight rollouts/question, temperature 1.0, and a 2,048-token cap. A one-/two-RTX-5090 pilot fixes a common budget: provisionally 6M generated tokens/run, four groups, three paired seeds (72M tokens). Report actual updates, prompt counts, length, truncation, and total GPU hours including all data generation, adaptation, scoring, training, and evaluation.

Policy evaluation. Evaluate the initial policy and final fixed-budget checkpoints with identical greedy decoding, without PRM reranking or voting. **D–C MATH500 accuracy is the primary comparison**; report per-seed differences, mean/std, and 95% paired question-bootstrap intervals. D vs. A/B tests practical value; small or inconsistent gains remain inconclusive.

Reward diagnostic. On fixed ProcessBench MATH solutions, report first-error accuracy, correct-chain acceptance, and their harmonic mean at threshold 0.5. Within answer-correct solutions, valid-chain AUROC tests discrimination beyond a global score shift. Report overlaps and class counts; no diagnostic is used for tuning. Reward hacking and policy distribution shift remain risks.

5 Three-person plan and tentative timeline

Hongyi Ding leads GRPO and GPU profiling; Jasmine Jiang leads PRM data and adaptation; Yang Zeng leads split audits, evaluation, and statistics. All three write the report.

Period	Deliverable and responsibility
Week 1	Hongyi: 48-hour resource/loss pilot; Jasmine: buffer and PRMs; Yang: split audit. Proposal due Oct. 21, 11:59 pm PT.
Week 2	Hongyi/Jasmine: first-seed A–D runs; Yang: verify rewards, accuracy, and cost logs.
Week 3	Hongyi/Jasmine: remaining seeds; Yang: paired comparisons and error analysis.
Week 4	All: final evaluation, reproducible code, figures, and report.

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